



BAULKHAM HILLS HIGH SCHOOL

2015

YEAR 12 HSC ASSESSMENT TASK 1

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 50 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions

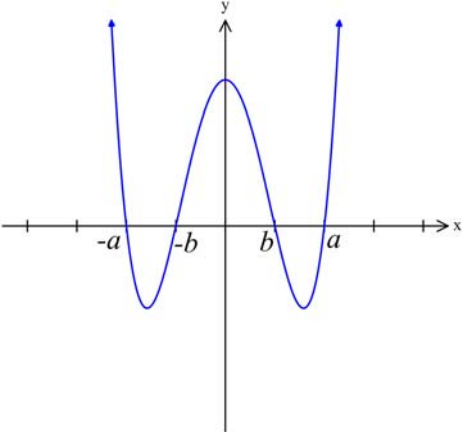
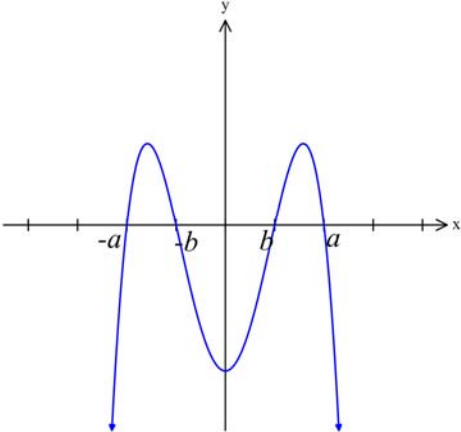
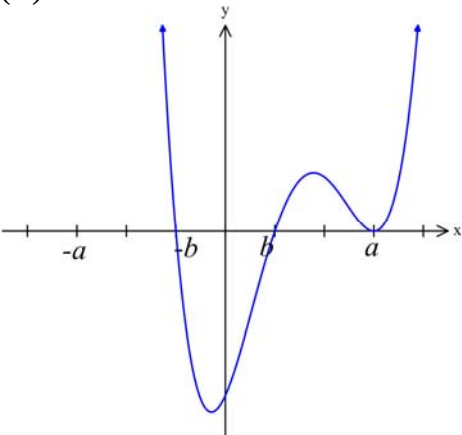
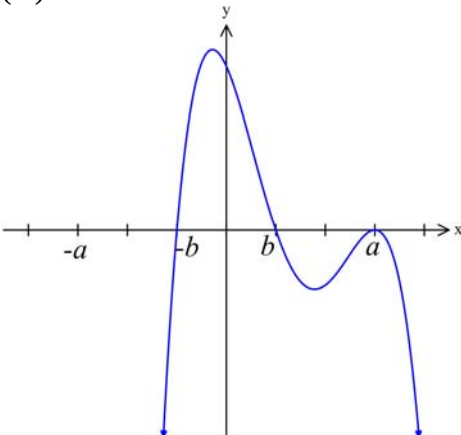
Total marks – 36

**Questions 1-3 – Extended Response
36 marks**

Answer each question on the appropriate pages of your answer booklet.

Question 1 (12 marks) - Answer on the appropriate page of your answer booklet		Marks
a)	If $2x - 3$ is a factor of $3x^3 + 3x + a$, find the value of a .	1
b)	A polynomial $P(x) = ax^n(2x - 5)^2 + 3x - 4$ is monic and has degree 8. (i) Find the value of a. (ii) Find the value of n. (iii) What is the remainder when $P(x)$ is divided by x?	1 1 1
c)	If $P(x) = 2x^3 + 11x^2 - 4$ and $A(x) = x + 1$, find the quotient and remainder when $P(x)$ is divided by $A(x)$.	2
d)	Without using calculus, sketch the graph of $y = (x - 1)^3(3 - x)$, showing the intercepts with the coordinate axes.	3
e)	Use mathematical induction to prove that $\frac{2}{1 \times 3} + \frac{2}{3 \times 5} + \frac{2}{5 \times 7} + \cdots + \frac{2}{(2n - 1)(2n + 1)} = 1 - \frac{1}{2n + 1}$ for all positive integers n.	3
End of Question 1		

Question 2 (12 marks) - Answer on the appropriate page of your answer booklet		
a)	If α, β and γ are the roots of $x^3 + 2x^2 - 3x - 5 = 0$, find the value of: (i) $\alpha + \beta + \gamma$ (ii) $\alpha\beta + \beta\gamma + \gamma\alpha$ (iii) $\alpha\beta\gamma$ (iv) $\frac{\gamma}{\alpha\beta} + \frac{\alpha}{\beta\gamma} + \frac{\beta}{\gamma\alpha}$	1 1 1 2
b)	Solve $2x^3 - 3x^2 - 8x + 12 = 0$.	3
c)	Millie is proving a statement by Mathematical Induction and has obtained the expression: $\frac{1}{6}k(k+1)(2k+1) + (k+1)^2$ Which of the following expressions is equivalent to the above expression? (A) $\frac{(k+1)(k+2)(k+3)}{6}$ (B) $\frac{(k+1)(k+2)(2k+3)}{6}$ (C) $\frac{(k+1)(2k+2)(k+3)}{6}$ (D) $\frac{(2k+1)(k+2)(k+3)}{6}$	1
d)	Use mathematical induction to prove $3^{2n} - 1$ is divisible by 8 for all positive integers n .	3
End of Question 2		

Question 3 (12 marks) - Answer on the appropriate page of your answer booklet	Marks
a) If $a > b$, the graph of $y = (x - a)^2(b^2 - x^2)$ could be: (A)  (B)  (C)  (D) 	1
b) $P(x)$ and $Q(x)$ are two polynomials such that $P(x) = (2x^2 + x + 3)Q(x) + 4x - 1$. When $Q(x)$ is divided by $x + 2$, the remainder is 1. Show that $x + 2$ is a factor of $P(x)$.	3
c) (i) Sketch the graph of $y = x^2(x - 1)(x - 2)$, showing the x-intercepts. (ii) Hence, or otherwise, solve the inequality: $\frac{x^2}{(x - 1)(x - 2)} \leq 0$	2 3
d) $P(x)$ is a polynomial of degree 3. Find $P(x)$, given that : $y = P(x)$ is an odd function. $(x - 2)$ is a factor of $P(x)$. - when $P(x)$ is divided by $(x + 4)$, the remainder is 96.	3

End of Examination

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Q1.

a) $P\left(\frac{3}{2}\right) = 0 \therefore 3\left(\frac{3}{2}\right)^3 + 3\left(\frac{3}{2}\right) + a = 0$

$$\frac{81}{8} + \frac{9}{2} + a = 0$$

$$a = -\frac{117}{8} \quad \leftarrow (1)$$

b) (i) $a = \frac{1}{4} \quad \leftarrow (1)$

(ii) $n = 6 \quad \leftarrow (1)$

(iii) Rem. = $P(0) = 0 + 0 - \frac{4}{3} = -\frac{4}{3} \quad \leftarrow (1)$

c) $\frac{2x^2 + 9x - 9}{x+1} \quad \leftarrow \text{quotient (1)}$

$$\begin{array}{r} 2x^2 + 9x - 9 \\ x+1 \overline{) 2x^3 + 11x^2 - 4} \end{array}$$

$$\underline{2x^3 + 2x^2}$$

$$9x^2$$

$$\underline{9x^2 + 9x}$$

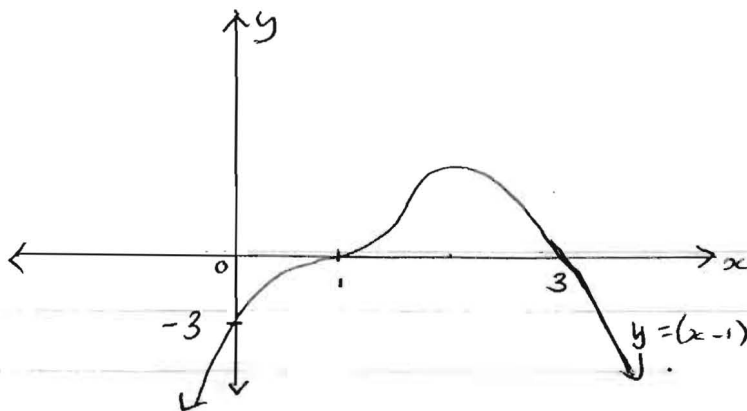
$$-9x - 4$$

$$\underline{-9x - 9}$$

$$5$$

$\leftarrow$ remainder (1)

d)



Single root $x=3$, (1)

Triple root $x=1$ (1)

y-int (1)

$[-1 \text{ if ends incorrect}]$

e) If $n=1$: $LHS = \frac{2}{(2-1)(2+1)} = \frac{2}{1 \times 3} = \frac{2}{3}$

$$RHS = 1 - \frac{1}{2+1} = 1 - \frac{1}{3} = \frac{2}{3}$$

$$LHS = RHS \therefore \text{True for } n=1$$

(1)

Assume true for $n=k$

i.e. Assume $\frac{2}{1 \times 3} + \frac{2}{3 \times 5} + \frac{2}{5 \times 7} + \dots + \frac{2}{(2k-1)(2k+1)} = 1 - \frac{1}{2k+1}$

Now prove true for $n=k+1$

i. Prove

$$\frac{2}{1 \times 3} + \frac{2}{3 \times 5} + \frac{2}{5 \times 7} + \dots + \frac{2}{(2k-1)(2k+1)} + \frac{2}{(2k+1)(2k+3)} = 1 - \frac{1}{2k+3}$$

LHS =

$$1 - \frac{1}{2k+1} + \frac{2}{(2k+1)(2k+3)} \quad \text{by (i) assumption}$$

$$= 1 - \left(\frac{1}{2k+1} - \frac{2}{(2k+1)(2k+3)} \right)$$

$$= 1 - \frac{2k+3 - 2}{(2k+1)(2k+3)}$$

$$= 1 - \frac{2k+1}{(2k+1)(2k+3)}$$

$$= 1 - \frac{1}{2k+3}$$

$$= \text{RHS} \quad \therefore \text{If true for } n=k, \text{ then also true for } n=k+1 \quad (1)$$

Now statement is true for $n=1$

$\therefore$ Also true for $n=2, 3, 4, \dots$ and by induction true for all positive integers n .

Q2.

a) i) $\alpha + \beta + \gamma = -2$ — (1)

ii) $\alpha\beta + \beta\gamma + \gamma\alpha = 3$ — (1)

iii) $\alpha\beta\gamma = 5$ — (1)

iv) $\frac{\gamma}{\alpha\beta} + \frac{\alpha}{\beta\gamma} + \frac{\beta}{\gamma\alpha} = \frac{\gamma^2 + \alpha^2 + \beta^2}{\alpha\beta\gamma}$

$$= \frac{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)}{\alpha\beta\gamma} \quad \text{— (1)}$$

$$= \frac{(-2)^2 - 2(3)}{5}$$

$$= 2. \quad \text{— (1)}$$

b) $2x^3 - 3x^2 - 8x + 12 = 0$

$P(1) \neq 0$

$P(-1) \neq 0$

$P(2) = 16 - 12 - 16 + 12 = 0 \therefore x - 2$ is a factor $\leftarrow$ (1)

$P(x) = (x - 2)(2x^2 + x - 6)$ $\leftarrow$ by long div. (1)
or otherwise

$$= (x - 2)(2x - 3)(x + 2)$$

$$= 0 \quad \text{when} \quad \underline{x = 2, -2, \frac{3}{2}} \quad \leftarrow (1)$$

ALT :

$$P(x) = x^2(2x - 3) - 4(2x - 3) \quad \leftarrow (1)$$

$$= (x^2 - 4)(2x - 3)$$

$$= (x - 2)(x + 2)(2x - 3) \quad \leftarrow (1)$$

$$\therefore P(x) = 0 \quad \text{when} \quad \underline{x = 2, -2, \frac{3}{2}} \quad \leftarrow (1)$$

$$\begin{aligned}
 c) \quad & \frac{1}{6} (k+1) (k(2k+1) + 6(k+1)) \\
 &= \frac{1}{6} (k+1) (2k^2 + 7k + 6) \\
 &= \frac{1}{6} (k+1) (2k+3) (k+2) \quad \therefore \textcircled{B} \quad \longleftarrow (1)
 \end{aligned}$$

d) If $n=1$ $3^{2n} - 1 = 3^2 - 1 = 8$ which is divisible by 8
 $\therefore$ True for $n=1$

Assume true for $n=k$

i.e. Assume $3^{2k} - 1 = 8m$ where m is an integer.

Now prove true for $n=k+1$

i.e. Prove $3^{2(k+1)} - 1 = 8p$ where p is an integer.

Now

$$\begin{aligned}
 3^{2(k+1)} - 1 &= 3^{2k+2} - 1 \\
 &= 3^2 \cdot 3^{2k} - 1
 \end{aligned}$$

$$= 9(3^{2k}) - 1$$

$$= 9(8m+1) - 1 \quad \text{by assumption}$$

$$= 8(9m) + 9 - 1$$

$$= 8(9m) + 8$$

$$= 8(9m+1)$$

$$= 8p \quad \text{where } p=9m+1 \text{ is an integer}$$

$\therefore$ If true for $n=k$, then also true for $n=k+1$.

Now statement is true for $n=1$

$\therefore$ Also true for $n=2, 3, 4, \dots$ and by induction true for all positive integers n .

Q3.

- a) Double root at $x=a$
Single root at $x=b, -b$.

Ends:

↓ ↓ (D) ← (1)

- b) $Q(x) = (x+2) \cdot S(x) + 1$ ← (1)
by the division transform
where $S(x)$ is a polynomial

$$P(x) = (2x^2 + x + 3)((x+2) \cdot S(x) + 1) + 4x - 1 \quad \leftarrow (1)$$

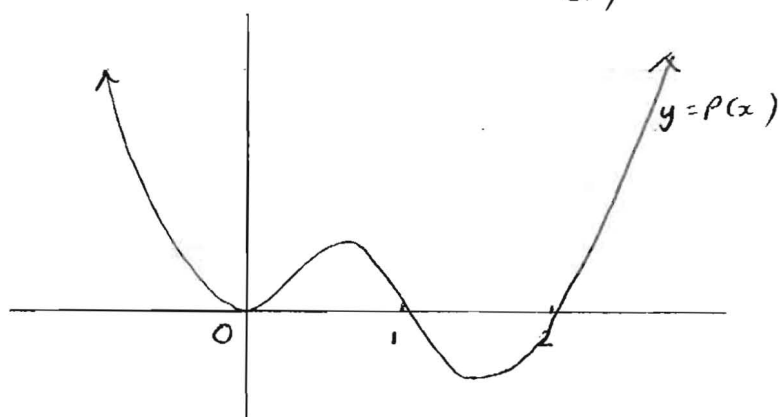
$$P(-2) = (8 - 2 + 3)(0 \cdot S(-2) + 1) + 4(-2) - 1$$

$$= 9 \times 1 - 8 - 1$$

$$= 0$$

← (1) $x+2$ is a factor of $P(x)$

c) (i)



Single roots at
 $x=1, 2$ (1)

Double root at
 $x=0$ (1)

(ii) Sign of $\frac{x^2}{(x-1)(x-2)}$ = Sign of $P(x)$ except
 $x \neq 1, 2$

$$\frac{x^2}{(x-1)(x-2)} \leq 0 \quad \text{when} \quad \frac{x=0}{\uparrow (1)}, \quad \frac{1 < x < 2}{\uparrow (2)}$$

[$1 \leq x \leq 2$ is only
1 mark]

$$d) \quad P(x) = ax^3 + bx$$

$$P(2) = 0 \quad : \quad 8a + 2b = 0$$

$$4a + b = 0 \quad \text{--- ①}$$

$$P(-4) = 96 \quad -64a - 4b = 96$$

$$16a + b = -24 \quad \text{--- ②}$$

← one or both
(i)

Solve simultaneously

$$12a = -24 \quad a = -2$$

$$-8 + b = 0 \quad b = 8$$

← one or both
(i)

$$\therefore P(x) = -2x^3 + 8x$$

← (i)